



HWA CHONG INSTITUTION
JC2 Preliminary Examinations
Higher 2

CANDIDATE NAME

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CT GROUP

17S7__

CENTRE NUMBER

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INDEX NUMBER

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BIOLOGY

9744/02

Paper 2 Structured Questions

27 August 2018

No additional materials are required.

2 hours

INSTRUCTIONS TO CANDIDATES

There are **five** question booklets (**I** to **V**) to this paper. Write your **name**, **CT group**, **Centre number** and **index number** in the spaces provided at the top of this cover page and on the lines provided at the top of the cover pages of Booklets **II**, **III**, **IV** and **V**.

This paper contains **ten** structured questions. Answer **all** questions in the spaces provided on the question paper.

INFORMATION FOR CANDIDATES

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

You are reminded of the need for good English and clear presentation in your answers.

For Examiners' Use	
1	/ 8
2	/ 8
3	/ 9
4	/ 12
5	/ 11
6	/ 10
7	/ 12
8	/ 8
9	/ 12
10	/ 10
Total Mark	/ 100

This document consists of **27** printed pages and **7** blank pages.

BOOKLET I

QUESTION 1

Fig. 1.1 shows a goblet cell from the colon of a rat, which can be found scattered among the epithelial cells that line the intestinal tract. The goblet cell produces mucus that is subsequently packaged into mucigen granules before being secreted out of the cell.

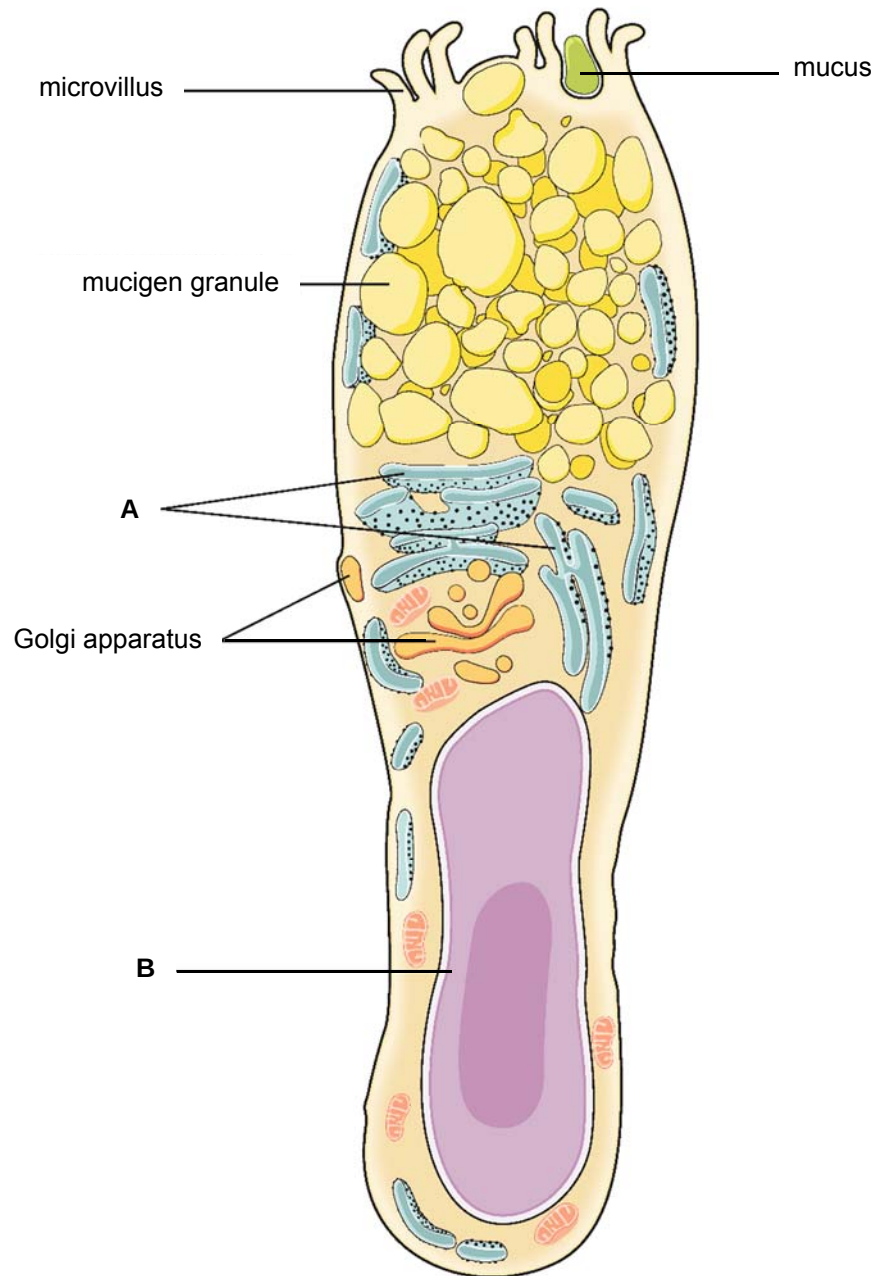


Fig. 1.1

(a) Name the organelles **A** and **B** shown in Fig. 1.1.

A :

B :

[2]

- (b) Explain how organelles **A** and **B** are adapted to allow the goblet cell to secrete mucus out of the cell.

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[4]

- (c) In 1967, Lynn Margulis proposed the theory of endosymbiosis, which states that mitochondria and chloroplasts of eukaryotes evolved from ancient prokaryotes. It proposed that these ancient prokaryotes were engulfed by other bacterial cells and both benefitted from the relationship – this has led to the evolution of eukaryotic cells.

Describe how the similarities between mitochondria and prokaryotic cells support the theory of endosymbiosis.

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[2]

[Total: 8]

QUESTION 2

Fig. 2.1 is a diagram representing the transport of glucose from the intestinal lumen to the extracellular fluid across an intestinal epithelial cell. Three proteins are labelled **P**, **Q** and **R**.

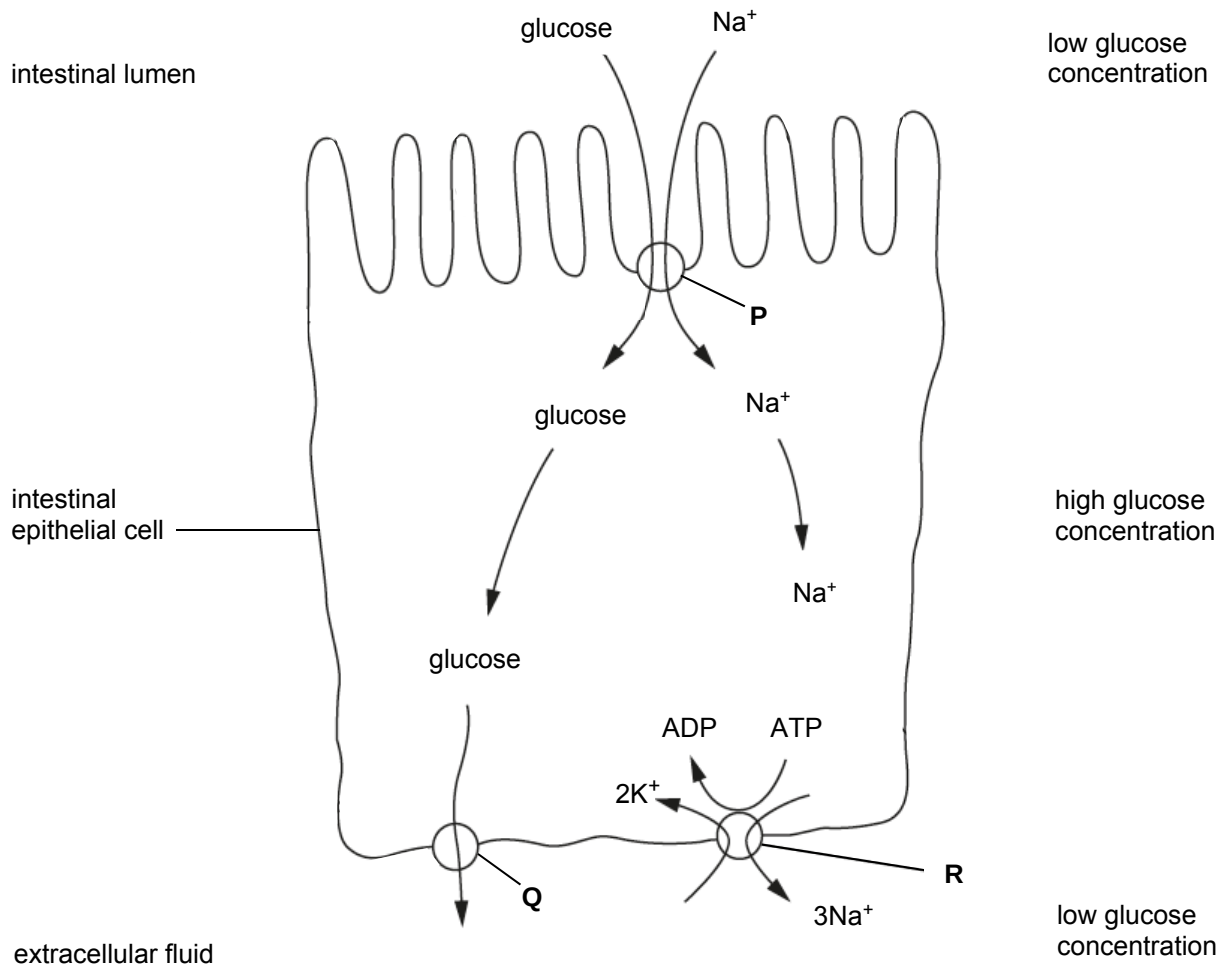


Fig. 2.1

- (a) Describe what is meant by *facilitated diffusion*.

[1]

- (b) Identify which transport protein, **P**, **Q** or **R**, is involved in facilitated diffusion.

[1]

(c) Explain how glucose is transported from the intestinal lumen to the extracellular fluid.

[4]

(d) Suggest why **R** is necessary in facilitating the transport of glucose across the epithelial cell in Fig. 2.1.

[2]

[Total: 8]

QUESTION 3

Fig. 3.1 shows the gene expression of a ribosomal protein, rS2.

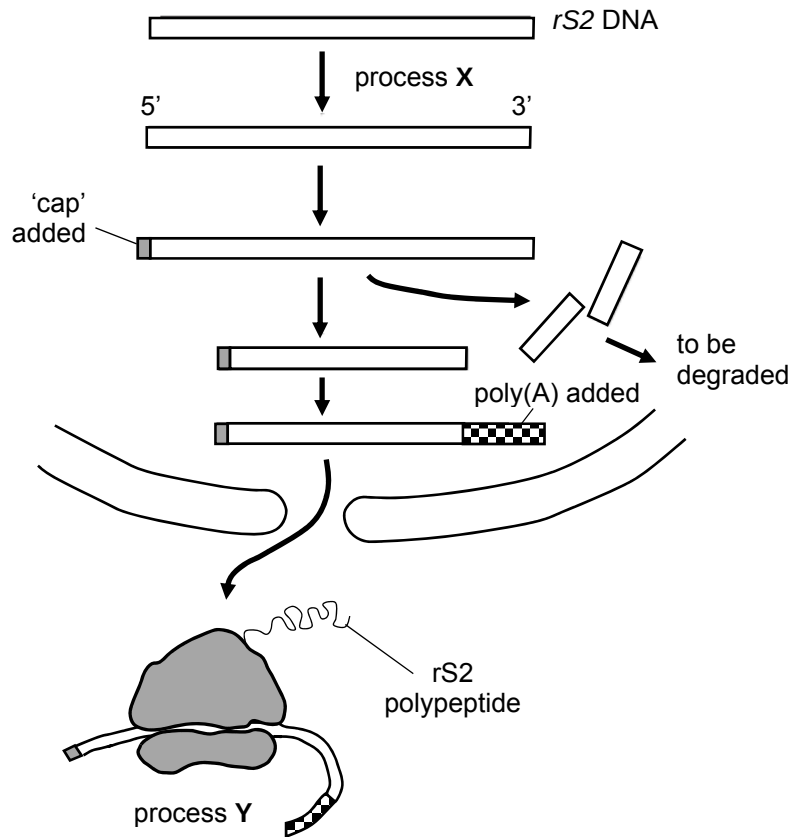


Fig. 3.1

- (a)** Describe the main structural features of a ribosome.

[2]

- (b)** State the role of the *rS2* promoter.

[1]

- (c) List three ways by which process X differs from process Y.

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..... [3]

- (d) Suggest how a mature ribosome is assembled upon the completion of process Y.

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[Total: 9]

QUESTION 4

Fig. 4.1 is an electron micrograph showing ribosomes involved in protein synthesis in an *Escherichia coli* cell.

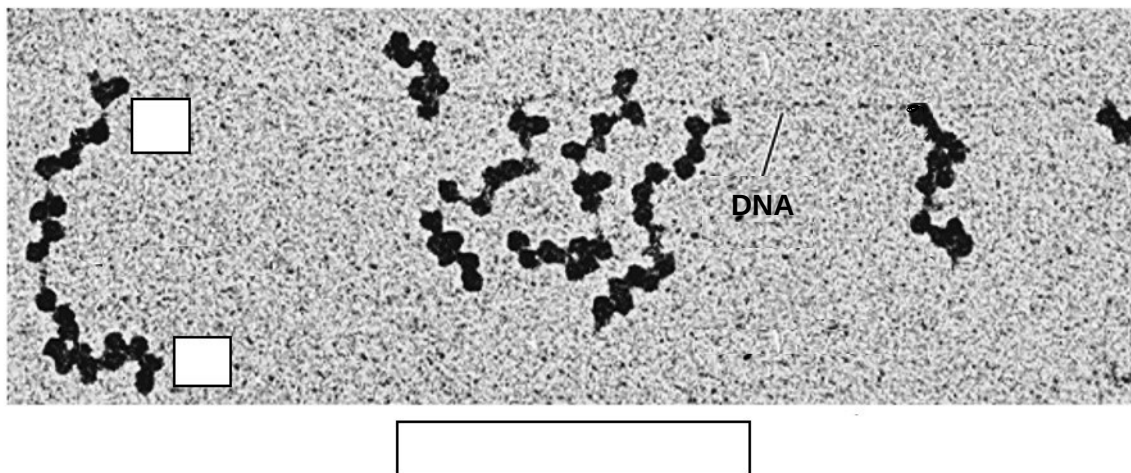


Fig. 4.1

(a)(i) On Fig. 4.1,

- label the 3' and 5' ends of the polynucleotide in the square boxes provided
- draw an arrow in the rectangular box provided to indicate the direction of transcription.

[2]

(ii) The mechanism of protein synthesis shown in Fig. 4.1 is unique to prokaryotes.

Explain why this is so.

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[4]

- (b) In *E. coli* cells, transcription is regulated by a cluster of genes known as an operon, which can be either inducible or repressible. The *trp* operon is one such operon.

(i) Explain if the *trp* operon is inducible or repressible.

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(ii) Describe what happens when tryptophan is absent in *E. coli*.

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..... [3]

- (c) *E. coli* was cultured in a growth medium with all the essential nutrients except tryptophan. Fig. 4.2 shows the changes in concentration of enzymes involved in tryptophan synthesis over time.

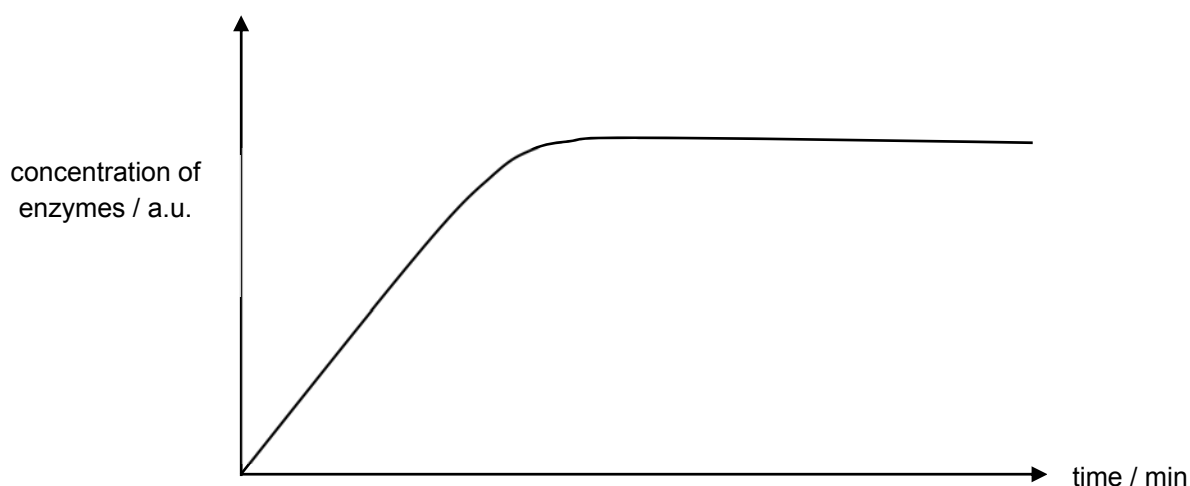


Fig. 4.2

As tryptophan is intrinsically synthesised, it is also utilised by *E. coli* at the same time.

Draw an arrow with the label X on Fig. 4.2 to indicate the time at which excess tryptophan was added to the growth medium. [1]

[Total :12]

QUESTION 5

Flooding is a severe threat for survival of terrestrial plants, mainly because it severely impedes the gas diffusion rate. Rice (*Oryza sativa*) is the only cereal that can be cultivated in the frequently flooded rivers.

An investigation was carried out into the effect of flooding on the growth of the submerged stems of rice plants. Young rice plants were grown in a container in which the level of water was increased in 10 cm steps, over a period of seven days. The mean length of the submerged internodes (lengths of stem between two leaves) and the concentration of ethylene in the rice stems was measured each day. As a control, rice plants were grown in identical conditions but the water level was kept constant throughout the seven days. The results are shown in Fig. 5.1.

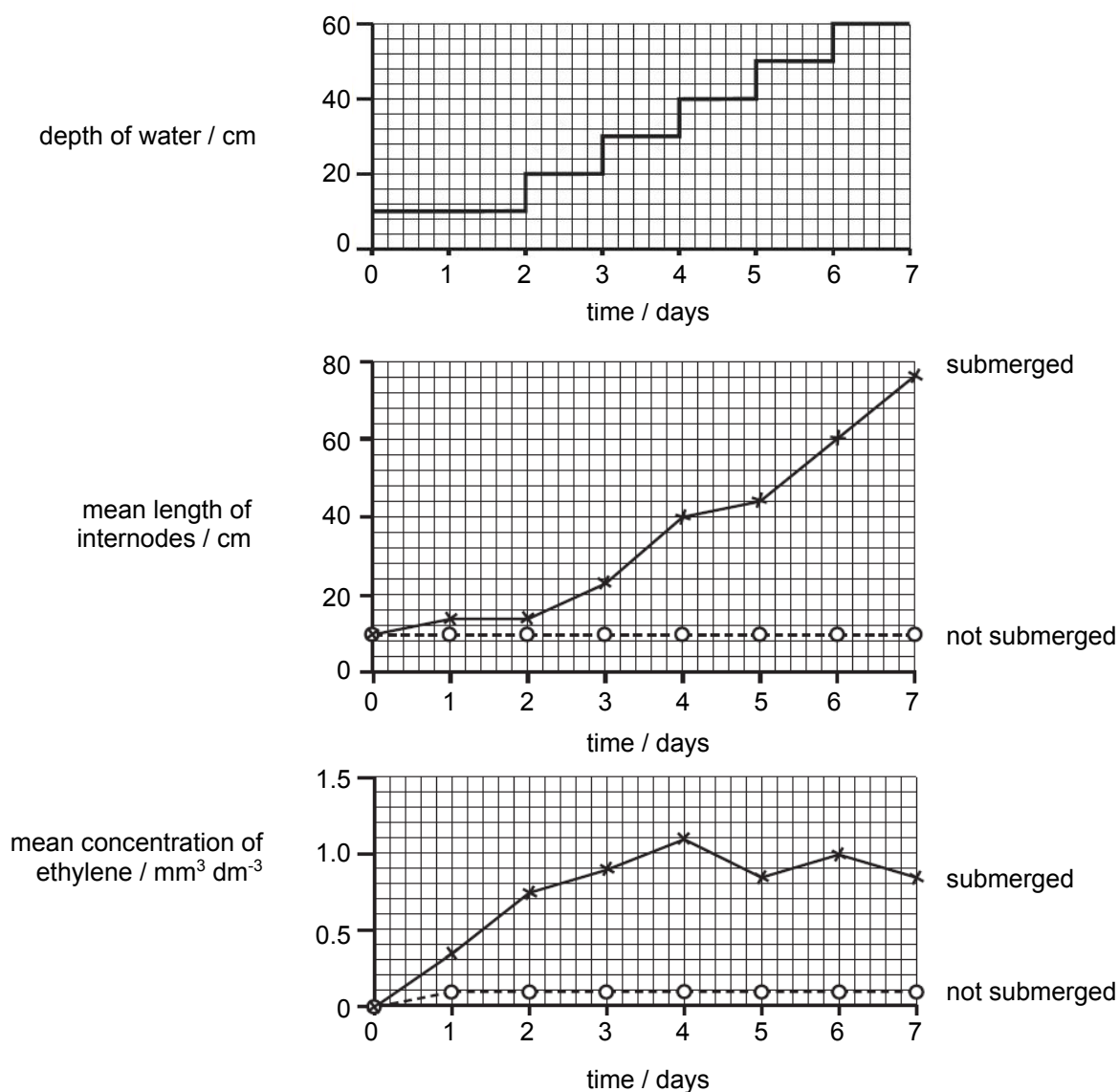


Fig. 5.1

- (a)** Describe the effects of increasing water level on the mean length of the submerged internodes and mean concentration of ethylene in rice stems.

[4]

- (b) Rice plants, when submerged for long periods of time, have evolved a survival strategy by increasing their internode lengths. The plant hormone that initiates this is gaseous ethylene. Submergence limits gaseous exchange, causing entrapment of gases and an increase in ethylene concentration within the cell.

Fig. 5.2 shows the effect of increased ethylene gas in cells of the rice plant. ETR1 and EIN2 are transmembrane proteins embedded in the membrane of the endoplasmic reticulum (ER).

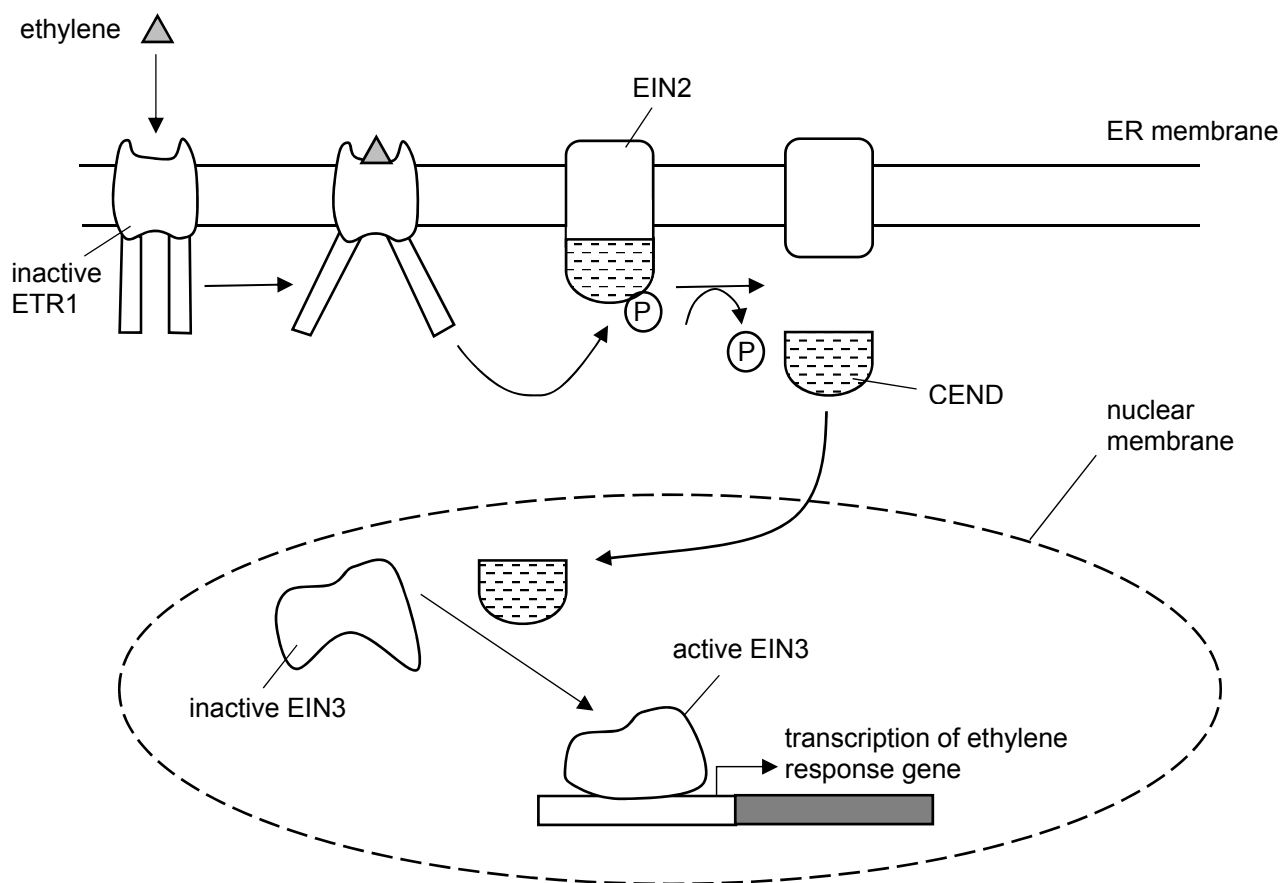


Fig. 5.2

- (i) Suggest how CEND is activated.

[2]

(ii) Explain how CEND increases the rate of transcription of the ethylene response gene.

[4]

(iii) Flooding is an indirect consequence of global warming.

Suggest how scientists can make use of these findings to mitigate the effects of flooding.

[1]

[Total: 11]

QUESTION 6

Actively dividing eukaryotic cells pass through a series of stages collectively known as the cell cycle.

- (a) Identify and describe what happens during the longest stage in a cell cycle.

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[3]

(b) A study on the synthesis and destruction of cyclins in dividing clam embryos was carried out, and consisted of the following steps.

- Radioactively labelled amino acid, ^{35}S -methionine was added to a suspension of fertilized clam eggs.
- The suspension was divided into two and colchicine was added to one.
- Duplicate samples were obtained at 5-minute intervals:
 - one to detect radioactivity in cyclin by autoradiography and
 - the other to identify stages of mitotic cell cycle by fixing and staining the cells.

The results are shown in Fig. 6.1.

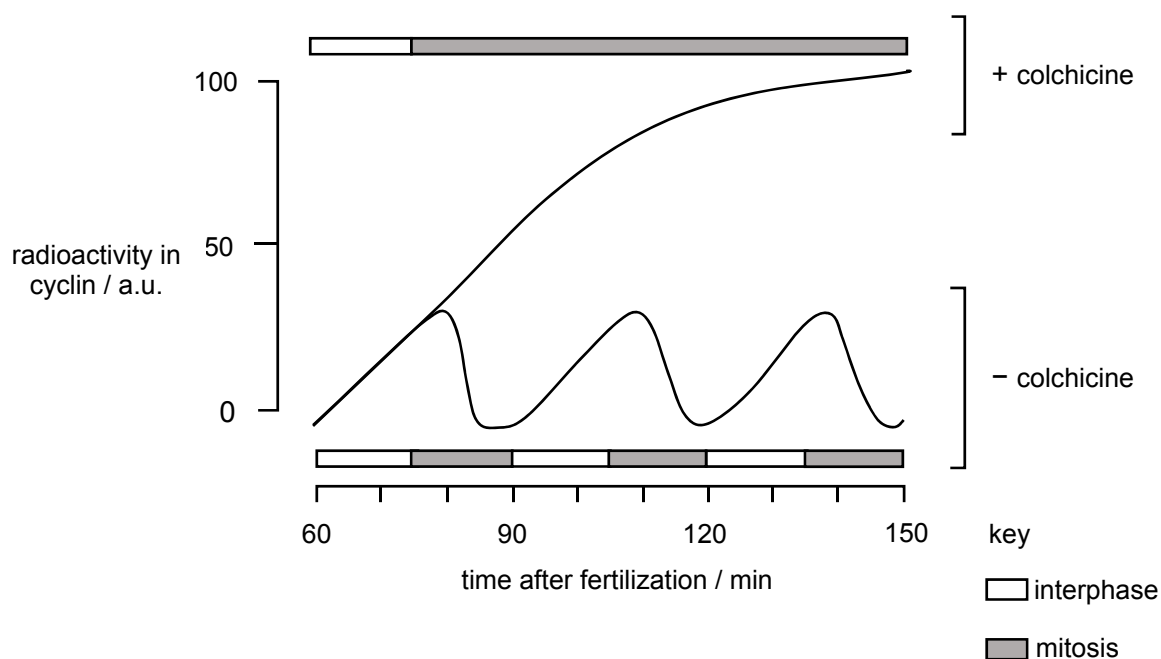


Fig. 6.1

Using the information shown in Fig. 6.1, describe and explain the effect of colchicine on mitosis.

[4]

(c) Research has shown that colchicine exerts its effect on tubulin molecules that make up microtubules. It is proposed that colchicine could be used to treat cancer.

(i) Explain how colchicine could be used for the treatment of cancer.

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..... [2]

(ii) Suggest a potential side effect of using colchicine in cancer treatment.

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..... [1]

[Total: 10]

QUESTION 7

Some pigeons have smooth heads, like their wild ancestors had. But on some domestic pigeons, the feathers on the back of the head and neck stand up to form a crest as shown in Fig 7.1. The gene controlling presence of crest is located on a different chromosome from the gene controlling plumage colour.

**Fig. 7.1**

When an ash-red female pigeon with no crest was crossed with an ash-red male pigeon with crest, all of the offspring had no crest. Table 7.1 shows the results of the cross.

Table 7.1

phenotype	number of offspring
ash-red male, no crest	7
ash-red female, no crest	3
blue female, no crest	4

Using the results shown in Table 7.1, a geneticist proposed a hypothesis which predicted a 3:1 ratio of ash-red:blue phenotype.

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Where Σ = sum of
 O = observed value
 E = expected value

(a)(i) Using the formula above, calculate the value for χ^2 . Show your working clearly.

- (ii) Given the critical value for χ^2 at $p = 0.05$ with one degree of freedom is 3.84, draw a conclusion and explain why this conclusion should be treated with caution.

[3]

- (b) In pigeons, presence of crest is autosomal and the allele for no crest (**D**) is dominant to crest (**d**).

Plumage colour, on the other hand, is sex-linked and controlled by three alleles. Ash-red allele (**A**) is dominant to blue (**B**), which is dominant to brown (**N**).

- In birds, the sex chromosomes are referred to as Z and W, rather than X and Y as in mammals. The W chromosome has no genes that affect plumage colour.
- The heterogametic sex is the female, not the male. Thus the male has two Z chromosomes (ZZ) and the female has one Z and one W chromosome (ZW).

Using the symbols for the alleles stated on page 23, draw a genetic diagram to explain the results shown in Table 7.1.

[5]

- (c) Suggest a method for the breeder to determine the genotype of a blue male pigeon with crest.

[2]

[Total: 12]

QUESTION 8

Chloroplasts belong to a group of organelles called plastids. Although different types of plastids have different structures and functions, one type of plastid can change into another type of plastid in response to environmental or developmental signals.

- Example 1: plants grown in the dark have plastids called etioplasts which lack chlorophyll. If these plants are exposed to light, the etioplasts quickly change into chloroplasts.
- Example 2: chloroplasts in surface tissues of tomato fruits change into plastids called chromoplasts as the fruits ripen. Thylakoid membranes break down and chlorophyll synthesis stops. Chromoplasts synthesise and accumulate red lycopene and orange β -carotene pigments.

- (a) For each of these examples, explain the effect on the rate of photosynthesis of one type of plastid changing into another type of plastid.

example 1

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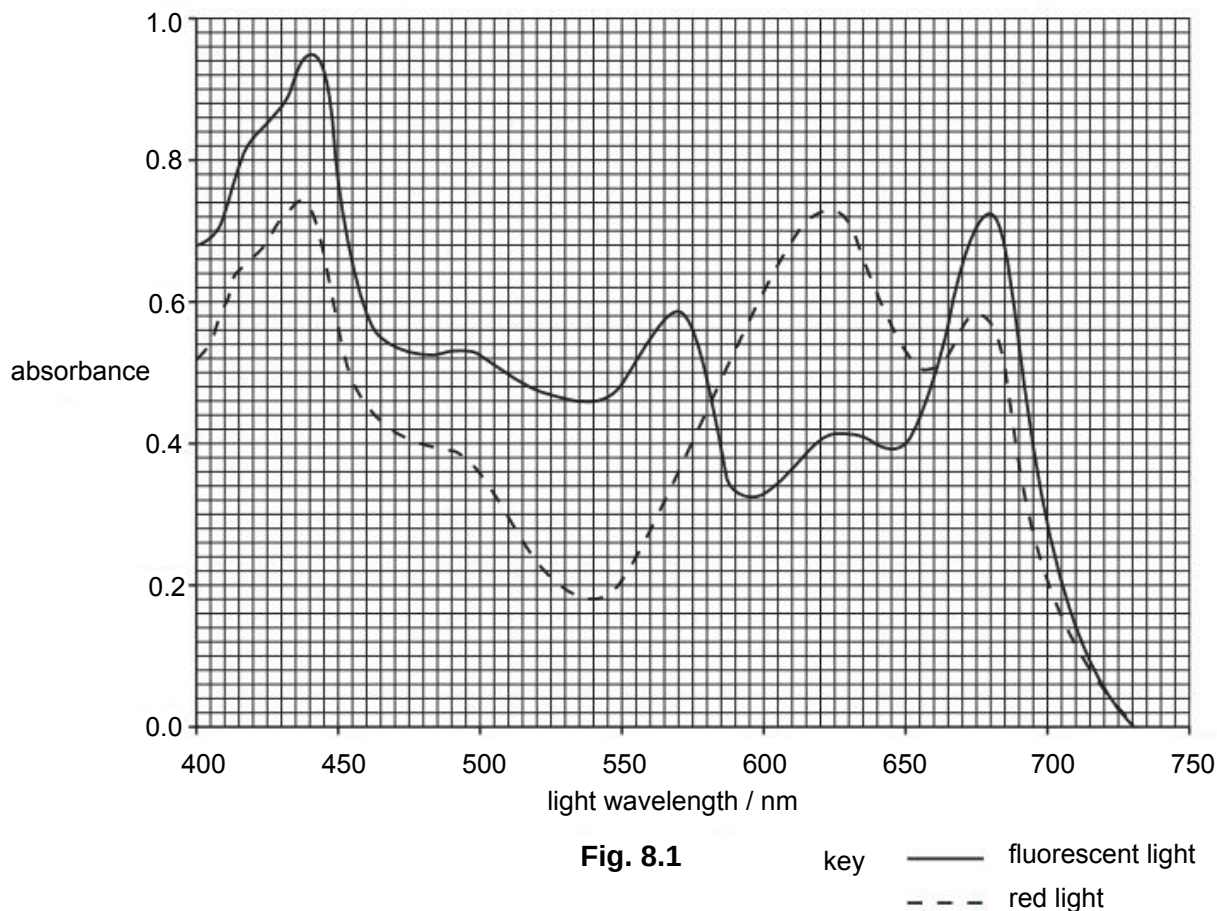
example 2

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..... [4]

- (b) Fig. 8.1 shows the absorption spectra of a type of photosynthetic bacteria, cyanobacteria, grown under two different lighting conditions. One group was grown under fluorescent light and the other group under red light. The range of light wavelengths absorbed by each group of cyanobacteria was then measured under identical lighting conditions.



- (i) Define the term *absorption spectrum* for photosynthesis.

[1]

- (ii) With reference to Fig. 8.1, explain the effect of different lighting conditions on the absorption spectra of the two groups of cyanobacteria.

[3]

[Total: 8]

(a) Suggest how the ability to use echolocation may have evolved from an ancestor that did not have that ability.

[4]

Table 9.1 shows data for both species.

species	mean body mass (g)	mean wingspan (m)	range of echolocation call (kHz)	colour
<i>P. pipistrellus</i>	5.5	0.22	42–47	medium to dark brown
<i>P. pygmaeus</i>	5.5	0.21	52–60	medium to dark brown

- (ii) Using the data in Table 9.1, suggest why pipistrelles were originally classified as one species.

[2]

Scientists have reconstructed phylogeny for both the *P. pipistrellus* and the *P. pygmaeus*. Fig. 9.1 shows the results of a phylogenetic analysis based on a 402 base pair portion of the *cytochrome b*, *cyt b*, gene.

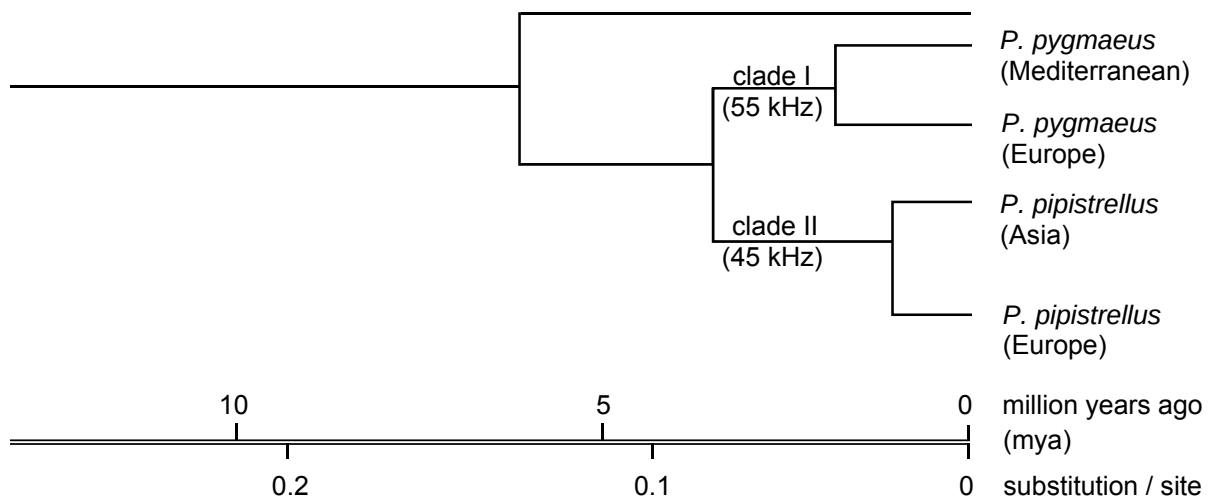


Fig. 9.1

- (c)(i) Using the information given in Table 9.1 and Fig. 9.1, discuss how both morphological and molecular data support the findings that *P. pipistrellus* and *P. pygmaeus* belong to different species.

[3]

- (ii) In more recent phylogenetic studies using other forms of molecular data, scientists have shown that *P. pipistrellus* and *P. pygmaeus* may have undergone convergent evolution in Europe, the Mediterranean and Asia.

Suggest how this is possible.

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..... [2]

[Total: 12]

QUESTION 10

The main cause of tuberculosis (TB) in humans is the bacterium, *Mycobacterium tuberculosis*.

Most cases of the disease involve the lungs. The bacterium can enter cells and remain inactive in a latent state. However, the bacterium can become active to produce symptoms of the disease.

In a person with active TB, the pathogen can be present in airborne droplets that are exhaled. Generally, a healthy person who inhales these droplets has effective defence mechanisms in the gas exchange system to prevent infection.

(a) One example of a defence mechanism against pathogens in the gas exchange system involves the action of macrophages.

(i) State the location in the body where macrophages have their origin.

..... [1]

(ii) Describe the mode of action of a macrophage.

.....

 [3]

(iii) It is sometimes possible for *M. tuberculosis* to survive within macrophages.

Suggest **one** way in which *M. tuberculosis* may survive within a macrophage.

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 [1]

- (b) The standard treatment for TB continues for six months and initially involves the use of four different antibiotics.

If no antibiotic resistance is detected, the treatment is reduced to two of the four antibiotics. The two antibiotics used are rifampicin and isoniazid.

Suggest the benefits of beginning the treatment with four different antibiotics.

[2]

Multidrug-resistant TB (MDR-TB) occurs if resistance develops to rifampicin and isoniazid.

The treatment for MDR-TB can last up to 30 months and involves different antibiotics to the standard treatment.

Table 10.1 shows the number of reported cases of TB and MDR-TB in the South-East Asia region between 2005 and 2014, as published by the World Health Organization (WHO).

Table 10.1

year	total number of reported cases of TB	total number of reported cases of MDF-TB
2005	1 947 603	68
2006	2 104 673	779
2007	2 202 149	918
2008	2 287 803	1717
2009	2 328 230	2560
2010	2 332 779	4263
2011	2 358 127	6615
2012	2 331 455	14 957
2013	2 297 033	18 384
2014	2 580 605	17 386

(c) TB is a disease of global importance.

Discuss the factors influencing the trends shown in Table 10.1.

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..... [3]

[Total: 10]

--- END OF PAPER ---

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